(ii) For every objects $X, Y \in Ob(\mathcal{C})$, the class $\mathrm{Hom}_{\mathcal{C}}(X, Y) = \mathrm{Hom}(X, Y)$ of morphisms (or arrows) from X, Y (for $f \in \mathrm{Hom}(X, Y)$, one may write $f : X \to Y$).

(iii) For any objects $X, Y, Z \in Ob(\mathcal{C})$, a composition map

$$\mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z), \quad (f, g) \mapsto f \circ g.$$

This data is required to satisfy the following axioms:

1. The composition is associative, i.e., $(f \circ g) \circ h = f \circ (g \circ h)$.

2. For each $X \in Ob(\mathcal{C})$, there is a morphism $1_X \in \mathrm{Hom}(X, X)$, called the **unit morphism**, such that $1_X \circ f = f$ and $g \circ 1_X = g$ for any f, g for which compositions make sense.

Remark 7.1.2. We will write $X \in \mathcal{C}$ instead of $X \in Ob(\mathcal{C})$.

Example 7.1.3. 1. The category **Sets** of sets (morphisms are arbitrary maps).

2. The categories **Groups**, **Rings** (morphisms are homomorphisms).

3. The category $\mathbf{Vect}_k$ of vector spaces over a field k (morphisms are linear maps).

4. The category $\mathrm{Rep}(A)$ of representations of an algebra A (morphisms are homomorphisms of representations). It is also denoted by $A - \mathrm{mod}$ (the category of left A-modules).

5. The category of topological spaces (morphisms are continuous maps).

6. The homotopy category of topological spaces (morphisms are homotopy classes of continuous maps).

Important remark. Unfortunately, one cannot simplify this definition by replacing the word "class" by the much more familiar word "set". Indeed, this would rule out the important Example 7.1.3(1), as it is well known that there is *no* set of all sets and that working with such a set leads to contradictions. The precise definition of a class and the precise distinction between a class and a set is the subject of set theory and cannot be discussed here. Luckily, for most practical purposes (in particular, in these notes) this distinction is not essential.